Comprehensive Model Results Summary

Hedonic Price Index Analysis

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14 Models: 7 Types × 2 Time Dimensions (Quarterly/Annual)

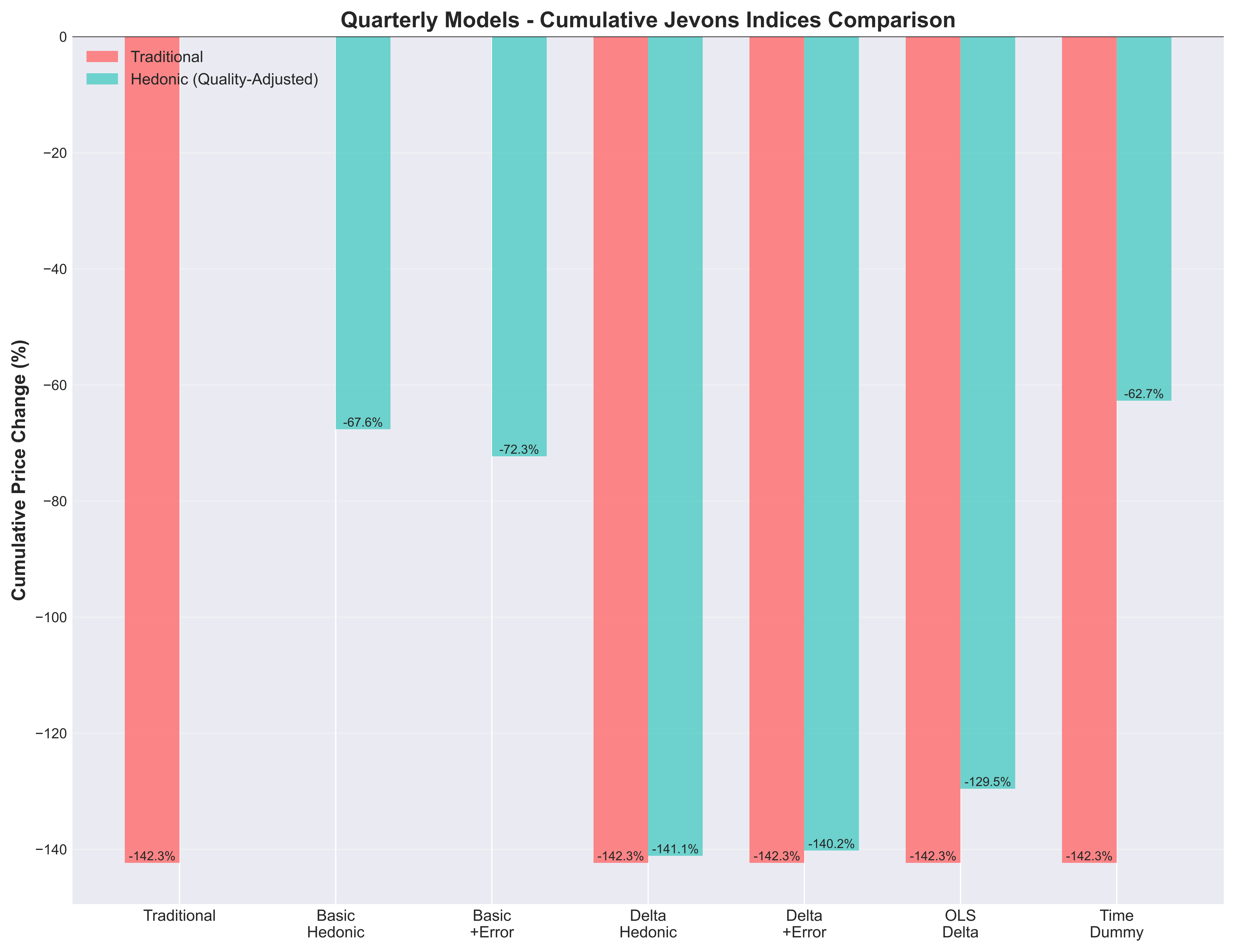
Quarterly Models - Cumulative Jevons Indices

Model	Traditional	Hedonic	Description
Traditional Jevons Index	-1.4231 (-142.31%)	N/A	No quality adjustment (2020 Q1+)
Basic Hedonic	N/A	-0.6759 (-67.59%)	Standard hedonic regression
Basic + Error Feature	N/A	-0.7226 (-72.26%)	With error feature
Delta Model	-1.4231 (-142.31%)	-1.4110 (-141.10%)	Direct price change modeling
Delta + Error	-1.4231 (-142.31%)	-1.4020 (-140.20%)	Delta with error feature
OLS Delta Model	-1.4231 (-142.31%)	-1.2952 (-129.52%)	OLS regression (no regularization)
Time Dummy Model	-1.4231 (-142.31%)	-0.6268 (-62.68%)	Pooled data + time dummy

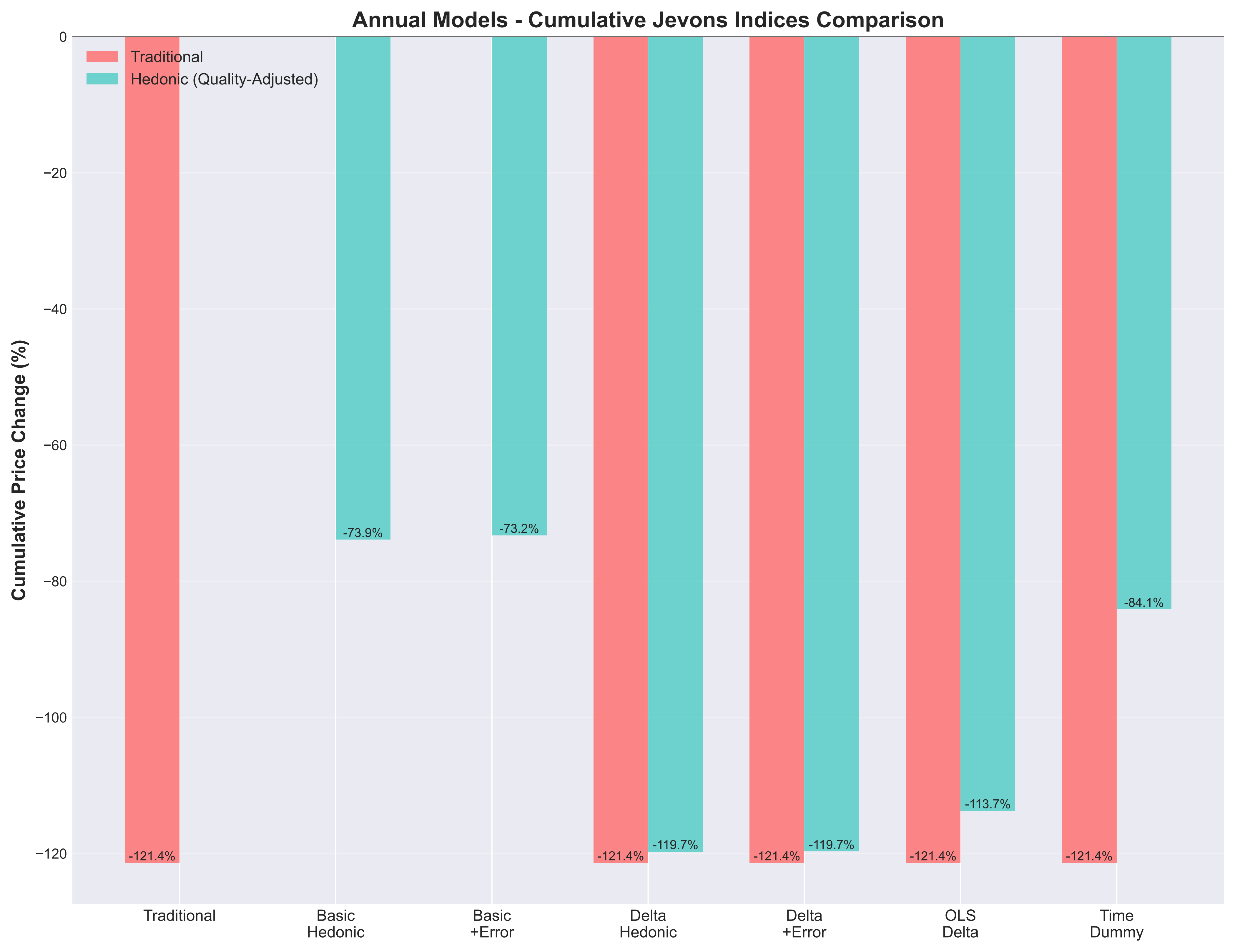
Annual Models - Cumulative Jevons Indices

All models cover 2020 to 2025 (5 periods)

Model	Traditional	Hedonic	Description
Traditional Jevons Index	-1.2135 (-121.35%)	N/A	No quality adjustment (2020+)
Basic Hedonic	N/A	-0.7387 (-73.87%)	Standard hedonic regression
Basic + Error Feature	N/A	-0.7325 (-73.25%)	With error feature
Delta Model	-1.2135 (-121.35%)	-1.1970 (-119.70%)	Direct price change modeling
Delta + Error	-1.2135 (-121.35%)	-1.1968 (-119.68%)	Delta with error feature
OLS Delta Model	-1.2135 (-121.35%)	-1.1371 (-113.71%)	OLS regression (no regularization)
Time Dummy Model	-1.2135 (-121.35%)	-0.8412 (-84.12%)	Pooled data + time dummy



Quarterly Models - Cumulative Jevons Indices Comparison
Traditional
Hedonic (Quality-Adjusted)
Cumulative Price Change (%)
0
-20
-40
-60
-80
-100
-120
-140
-142.3%
-67.6%
-72.3%
-142.3%
-141.1%
-142.3%
-140.2%
-142.3%
-129.5%
-142.3%
-62.7%
Traditional
Basic Hedonic
Basic +Error
Delta Hedonic
Delta +Error
OLS Delta
Time Dummy



Annual Models - Cumulative Jevons Indices Comparison
Traditional
Hedonic (Quality-Adjusted)
Cumulative Price Change (%)
0
-20
-40
-60
-80
-100
-120
-121.4%
-73.9%
-73.2%
-121.4%
-119.7%
-121.4%
-119.7%
-121.4%
-113.7%
-121.4%
-84.1%
Traditional
Basic Hedonic
Basic +Error
Delta Hedonic
Delta +Error
OLS Delta
Time Dummy

Model Descriptions

- **Traditional Jevons Index:**
  Direct calculation using actual prices. No quality adjustment. Serves as baseline.

- **Basic Hedonic:**
  Standard hedonic regression. Independent Lasso model for each period. Controls for product features.

- **Basic + Error Feature:**
  Hedonic regression with previous period prediction error as additional feature. Captures time dependencies.

- **Delta Model (Lasso):**
  Directly models log price differences between consecutive periods using Lasso. Uses OOF predictions to avoid mean-matching artifacts. Automatic feature selection.

- **Delta + Error Feature:**
  Combines Delta model with error feature from previous period pair. Captures both price changes and time dependencies.

- **OLS Delta Model:**
  Directly models log price differences using OLS (no regularization). All features included. Simpler interpretation.

- **Time Dummy Model:**
  Pools consecutive periods data together. Adds time dummy variable. Single model predicts both periods. More efficient parameter sharing.